

ELIZABETH CITY, NC, USA

WMO: 723070

Lat: 36.261N

Lon: 76.175W

Elev: 13

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 13786

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 21.4	(c) 25.3	(d) 5.8	(e) 7.4	(f) 26.4	(g) 10.0	(h) 9.2	(i) 30.1	(j) 26.9	(k) 38.2	(l) 24.7	(m) 41.5	(n) 9.5	(o) 290	(p) 0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.6	(c) 92.8	(d) 78.2	(e) 90.6	(f) 77.5	(g) 88.5	(h) 76.5	(i) 80.5	(j) 88.6	(k) 79.4	(l) 86.8	(m) 78.3	(n) 85.3	(o) 9.9	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.4	147.7	84.6	77.3	142.3	83.4	76.3	137.5	82.4	44.1	88.8	42.8	86.8	41.7	85.3	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	20.6	18.6	14.9	96.2	4.3	2.6	11.8	98.1	9.2	99.6	6.8	101.0	3.7	102.8
			12.9	82.7	4.1	1.7	10.0	84.0	7.7	85.0	5.4	86.0	2.4	87.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.7	42.3	45.1	50.9	60.4	68.9	76.5	79.5	78.7	73.7	63.7	53.4	46.7
	DBStd	15.29	10.55	9.21	9.31	8.33	7.13	5.11	4.11	3.91	5.22	8.04	8.39	9.59
	HDD50	834	286	191	105	10	0	0	0	0	0	4	57	181
	HDD65	2997	706	560	446	186	45	1	0	0	5	122	356	570
	CDD50	5124	47	53	133	323	587	796	914	890	712	430	160	79
	CDD65	1810	1	2	9	49	167	347	449	425	267	82	9	3
	CDH74	15382	3	11	61	356	1247	3068	4420	3866	1835	475	33	7
	CDH80	5150	0	0	6	72	350	1116	1690	1345	477	93	1	0
(14) Wind	WSAvg	9.0	9.5	10.0	10.6	10.7	9.5	8.5	7.8	7.3	8.6	8.5	8.6	8.8
(15) Precipitation	PrecAvg	47.8	3.9	3.4	3.8	3.0	4.1	4.4	5.7	5.3	4.5	3.5	2.9	3.1
	PrecMax	72.5	10.8	7.0	8.2	6.3	10.4	8.8	14.0	14.6	17.4	9.3	8.3	9.6
	PrecMin	36.1	1.3	1.0	1.0	0.6	0.6	1.2	0.8	1.2	0.2	0.0	0.6	0.3
	PrecStd	7.8	1.9	1.6	1.8	1.5	2.1	1.9	3.0	2.8	3.3	2.5	1.7	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.3	75.7	80.3	85.8	91.1	95.3	96.1	93.9	90.7	86.2	77.9	74.6
		MCWB	64.2	65.6	64.6	70.4	75.3	76.6	80.3	79.0	76.8	73.0	69.0	66.2
	2%	DB	67.5	69.8	74.8	81.9	86.8	91.8	92.8	91.3	87.7	82.5	73.5	69.9
		MCWB	61.7	60.4	62.5	67.6	73.1	76.1	79.5	78.3	75.5	71.5	65.7	64.9
	5%	DB	63.5	65.4	70.8	78.4	84.0	89.1	90.5	89.3	85.1	79.3	70.4	66.3
		MCWB	58.9	59.3	60.4	66.0	71.7	75.7	78.5	77.7	74.4	69.8	63.7	62.1
	10%	DB	59.2	61.2	66.4	74.4	81.0	86.4	88.1	87.0	82.5	76.0	66.9	62.9
		MCWB	54.5	54.6	59.3	64.2	70.2	74.4	77.4	76.4	73.3	68.5	61.8	58.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.3	67.0	67.7	73.8	77.5	80.2	83.0	81.9	79.8	77.1	71.9	68.8
		MCDB	70.5	72.9	75.6	81.1	85.8	88.5	91.9	90.6	86.2	81.2	74.5	71.8
	2%	WB	63.1	63.6	65.4	70.3	75.3	78.5	81.2	80.3	78.2	74.4	68.1	65.7
		MCDB	66.5	67.9	72.1	77.9	82.9	86.7	90.2	87.7	83.7	79.4	71.8	68.8
	5%	WB	59.4	59.9	63.0	68.1	73.5	77.4	79.8	79.1	76.9	72.5	65.4	62.9
		MCDB	62.9	64.7	68.3	75.5	80.6	85.7	88.1	85.7	82.2	77.1	68.8	65.9
	10%	WB	55.4	54.9	60.1	65.9	71.9	76.3	78.5	78.0	75.6	70.3	62.5	59.0
		MCDB	58.5	60.1	66.0	72.7	78.9	83.8	85.8	84.5	80.4	74.6	66.1	62.3
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	18.5	19.0	19.3	17.6	16.5	15.6	15.2	15.3	18.2	19.0	17.8
		MCDBR	22.6	23.9	24.1	23.2	20.0	19.3	17.8	17.3	17.6	19.7	20.8	19.9
		MCWBR	17.3	17.1	14.0	11.7	9.2	7.5	7.7	7.0	7.7	10.1	13.6	15.4
	5% WB	MCDBR	21.0	22.3	20.5	20.0	18.0	16.8	16.3	15.7	14.0	15.3	17.6	18.2
		MCWBR	18.0	17.7	14.7	11.8	9.0	7.5	7.8	7.1	7.0	9.0	13.7	15.8
(40) Clear-Sky Solar Irradiance	taub		0.312	0.325	0.351	0.394	0.438	0.490	0.516	0.505	0.413	0.361	0.328	0.318
	taud		2.535	2.492	2.448	2.337	2.245	2.152	2.090	2.109	2.370	2.499	2.538	2.549
	Ebn at Noon		282	290	291	282	270	255	247	247	267	274	273	272
	Edh at Noon		25	29	34	39	44	48	51	49	36	29	25	23
(44) All-Sky Solar Radiation	RadAvg		764	1004	1335	1706	1909	2023	1920	1735	1426	1150	859	684
(45)	RadStd		58	105	101	123	147	115	96	107	102	127	65	54

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+1.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+416	+293
Regional (53 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+202

Nomenclature: See separate page